

API Keys

Definition

An application programming interface (API) key is a secret unique identifier used to authenticate and authorize a user, developer, or calling program to an API.

Purpose

API keys serve as the primary mechanism for:

- **Authentication**: Verifying the identity of the caller
- **Authorization**: Granting access to specific resources or operations
- **Tracking**: Monitoring usage patterns and billing

Best Practices

Cloud computing providers such as Google Cloud Platform and Amazon Web Services recommend that API keys only be used to authenticate projects, rather than human users. This approach enhances security by separating project-level authentication from user-level credentials.

Security Considerations

- Keep API keys secret and never expose them in client-side code
- Rotate keys regularly to minimize risk of unauthorized access
- Use environment variables or secure key management systems
- Implement rate limiting to prevent abuse

Common Use Cases

- Third-party service integration
- Automated workflows and scripts

- Mobile application backend communication
- Webhook event handling

This document provides a foundational overview of API keys and their role in modern software development.